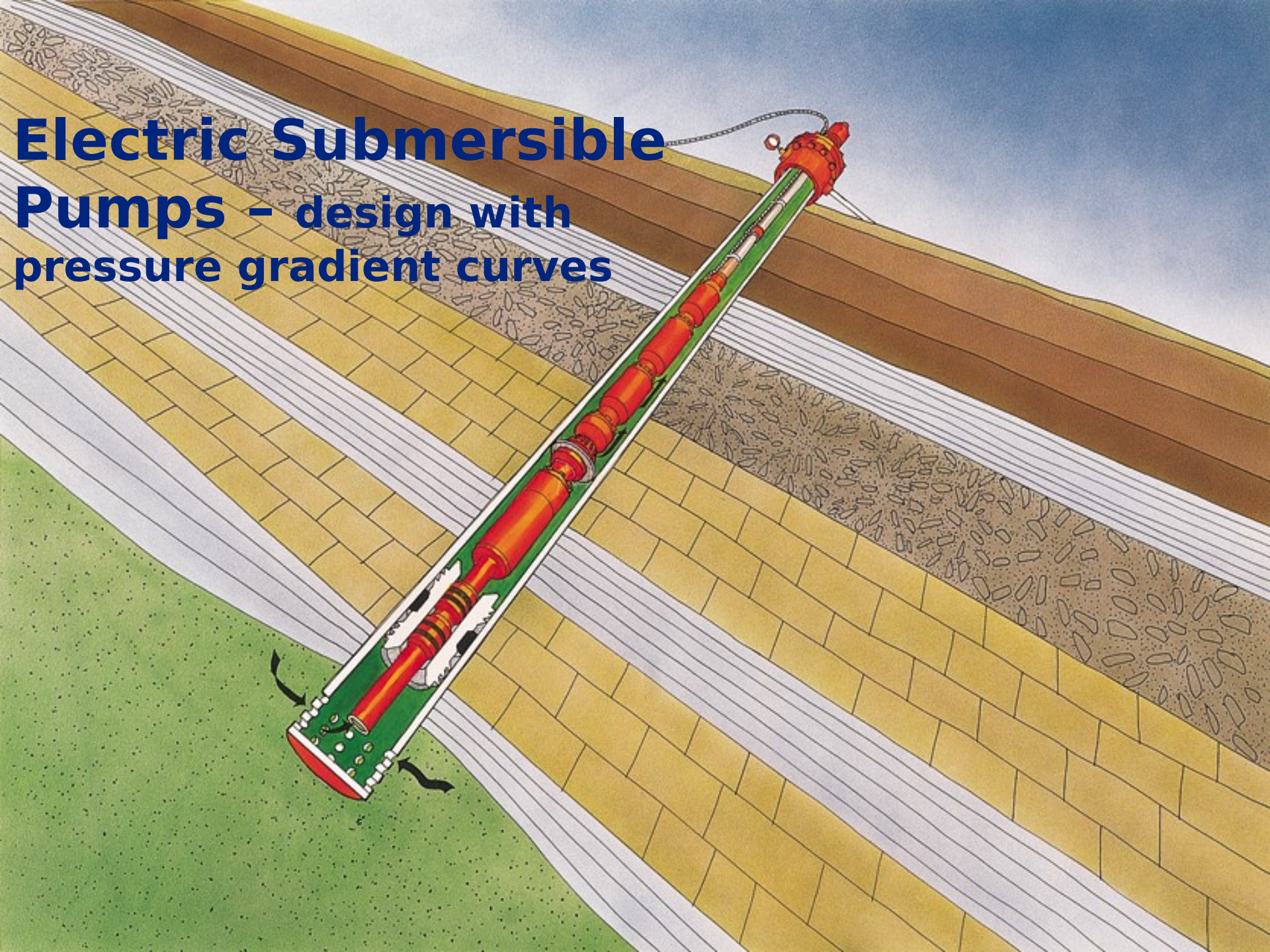
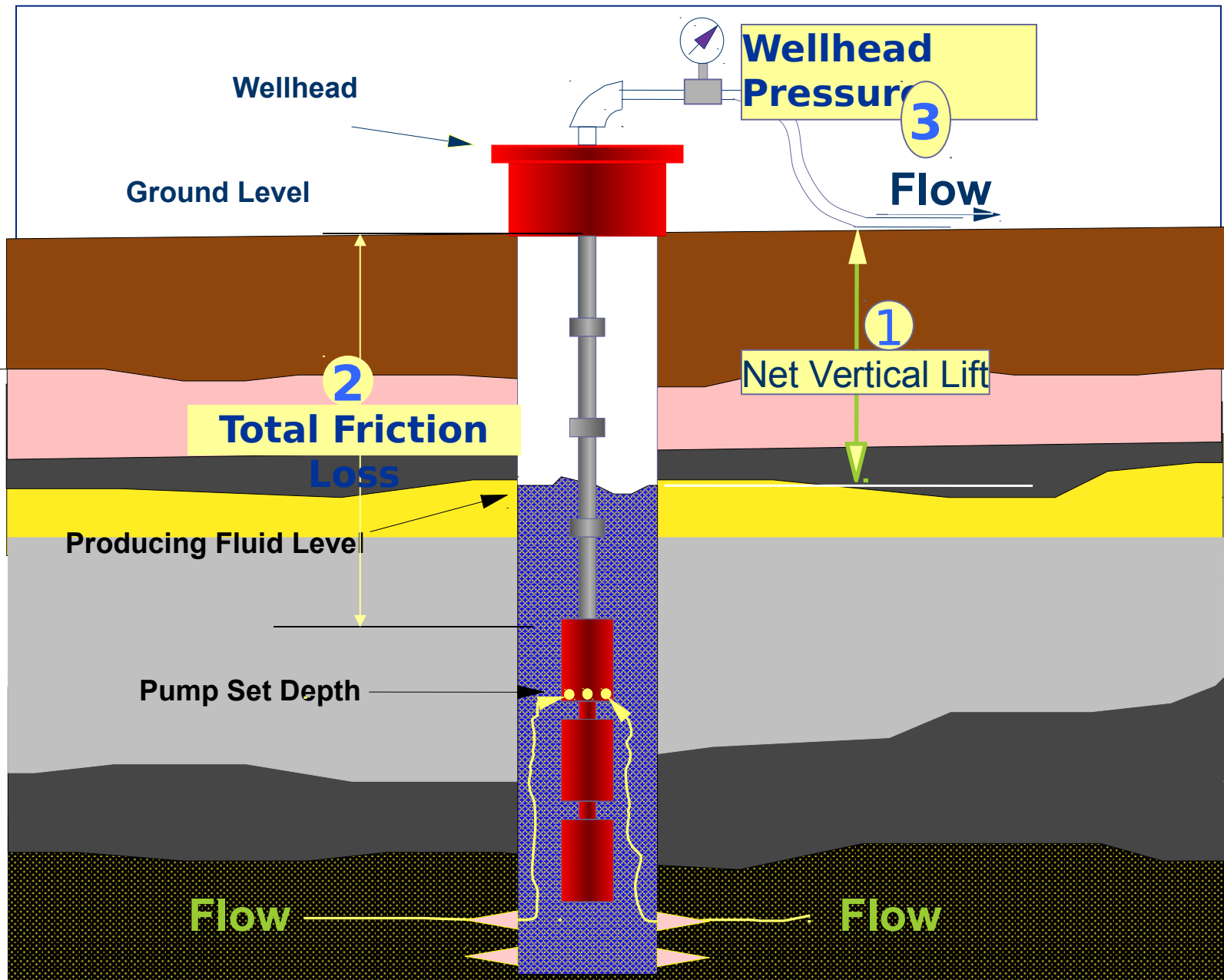


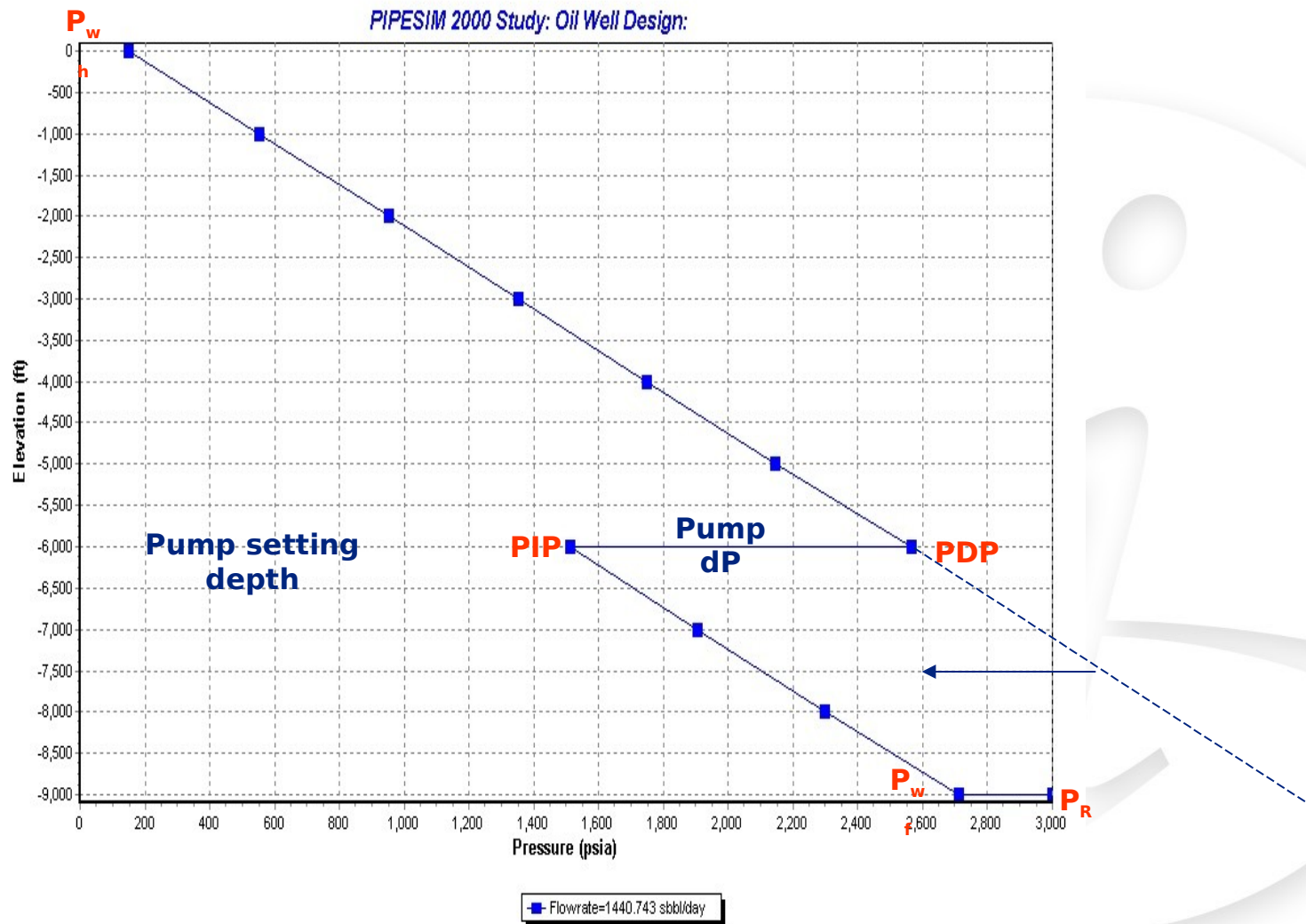
Electric Submersible Pumps – design with pressure gradient curves



Total Dynamic Head - What I am not going to talk about !



Pressure gradient curve

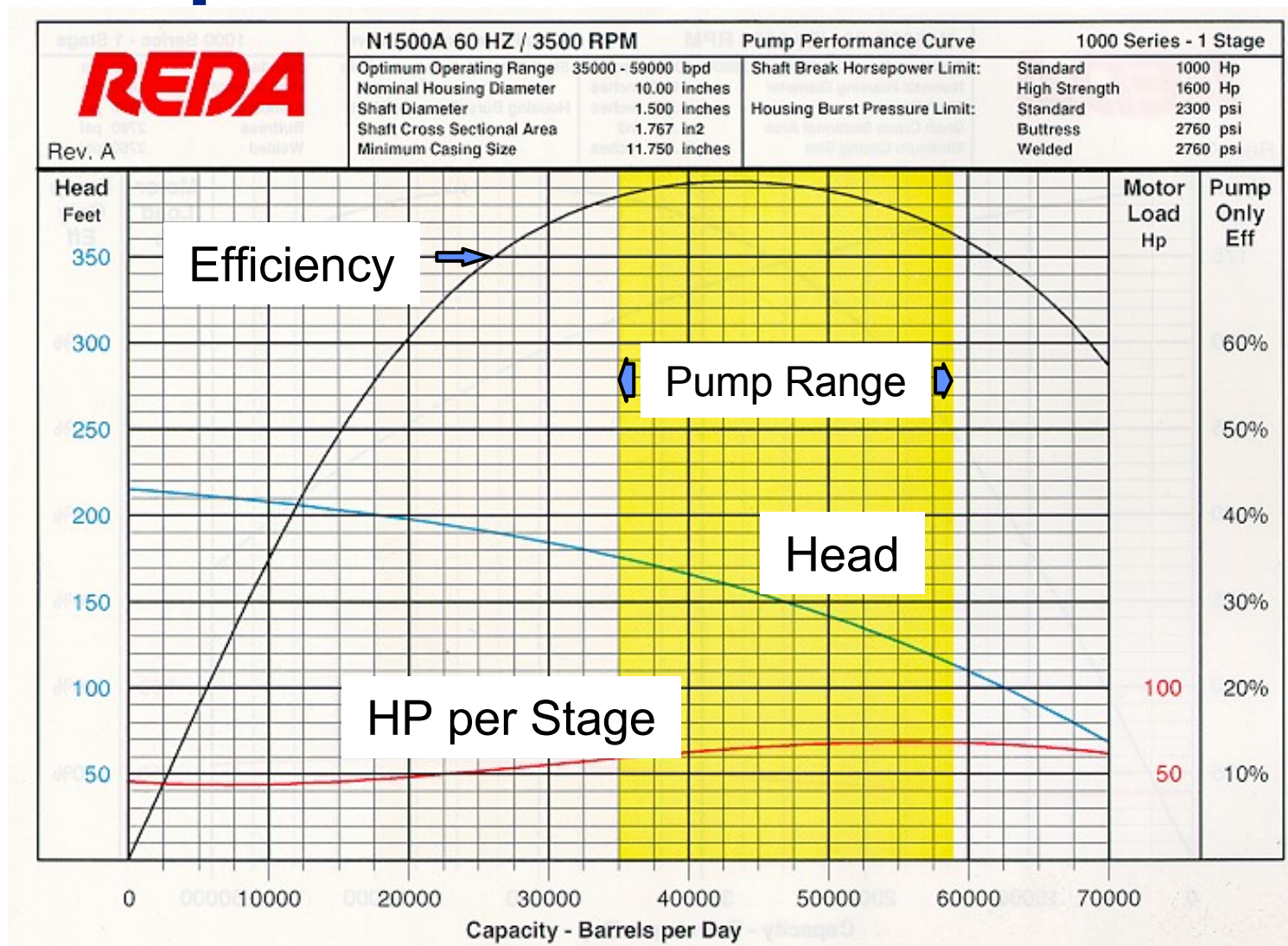


Baker Jandine
Created by EJA Demo on 15/02/02 13:45:45

Pump curves

- The two main pump design parameters are:
 - The pump delta P converted to head by fluid density.
 - The pump flow rate in rb/d

Pump curves



Systematic Pressure Analysis

- Above the pump:

$$P_{\text{discharge}} = \text{WHP} + \Delta P_{\text{gravity}} + \Delta P_{\text{friction}} \text{ (wellhead to pump)}$$

Validates : oil density, water cut, pressures and depths)

- Across the pump:

$$P_{\text{intake}} = P_{\text{discharge}} - \Delta P_{\text{pump}}$$

Where $\Delta P_{\text{pump}} = \text{Head (ft)} \times \text{Density (psi/ft)}$

Validates : pump performance, flowrate, effect of wear/viscosity/gas)

Systematic Pressure Analysis

- Below the pump:

$$P_{wf} = P_{intake} + \Delta P_{gravity} + \Delta P_{friction} \text{ (pump to reservoir)}$$

- In the reservoir:

$$PI = Q / (P_R - P_{wf}) \quad \text{(reservoir inflow performance)}$$

Validates : IPR, flowrate, PI, P_R

$$\text{Downhole flowrate (rb/d)} = Q_{oil}(\text{stbo/d}) \times B_o + Q_{water}(\text{stbw/d})$$

ESP Head/Pressure Calculations

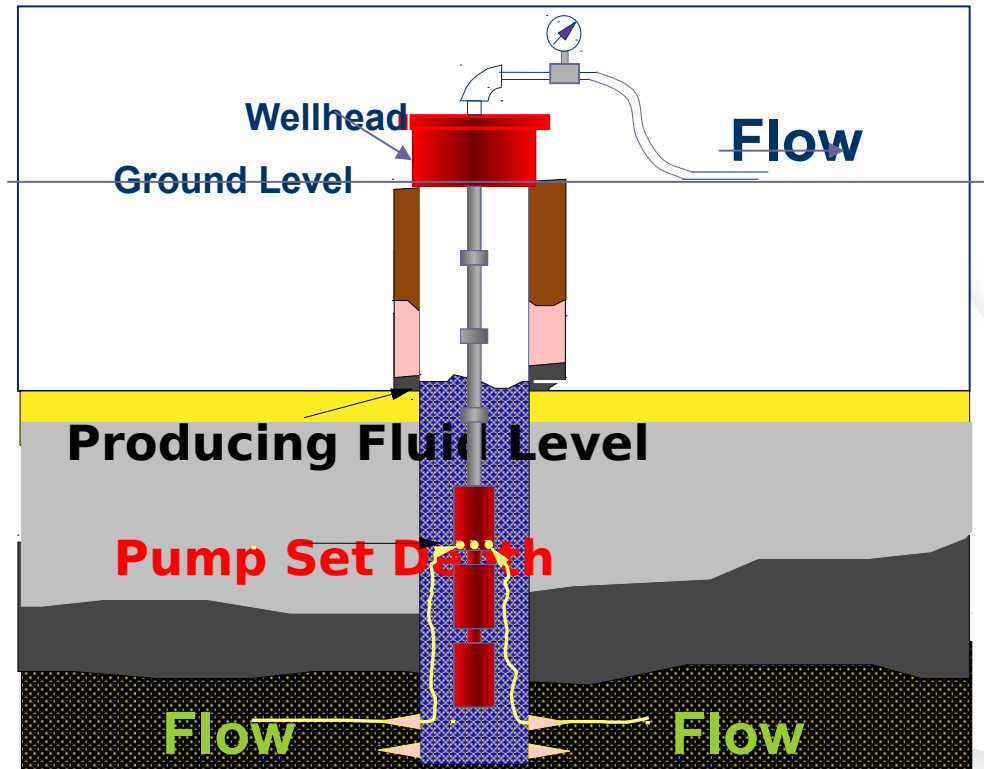
- ESP generates head by movement of fluids through the impeller and diffuser changing kinetic energy (movement) into potential energy (head)
- Head is dependent on
 - ✓ Speed of rotation of impeller
 - ✓ Flow rate through the impeller
 - ✓ Size and shape of impellers (flow types)
 - ✓ Viscosity of fluids
 - ✓ Amount of gas
 - ✓ Mechanical factors such as wear

ESP Head/Pressure Calculations

- Head (ft) = ΔP_{pump} (psi) / pressure gradient of well fluids (psi/ft)
 - remember pressure gradient (psi/ft) = S.G. / 2.31
- Questions:
 1. For a pump generating 2500 feet of head and pumping seawater, what is ΔP ?
 2. For the same pump which is now pumping 34°API oil what is ΔP ?
 3. What would this scenario be typical of ?

ESP Head/Pressure Calculations

- Fluid level above the pump

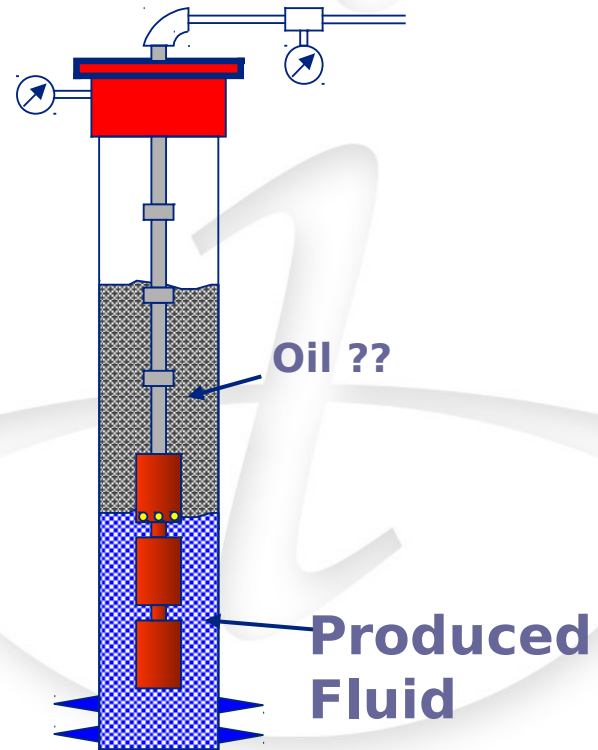


Given : pump set depth = 6000 ft, top of perforations = 6750 ft, the measured P_{intake} = 600 psi and zero casing pressure in the well producing 100% 34°API oil , what is the fluid level above the pump ?

ESP Head/Pressure Calculations

- For the case of this well with no measurement of P_{intake} but with 40% water cut and a measured fluid level= 4379 ft, what would be the P_{wf} ?

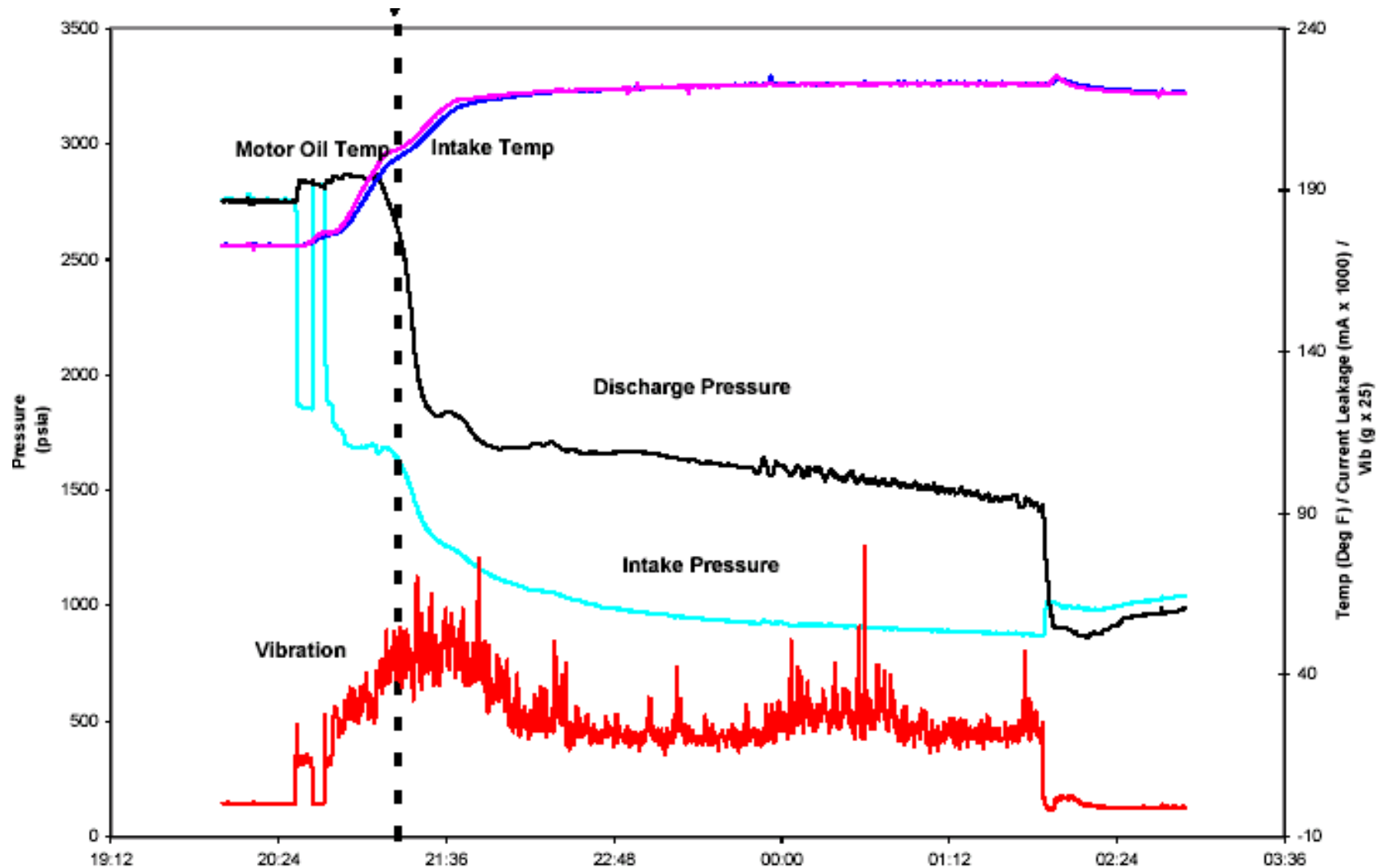
Downhole measurement will reduce uncertainty



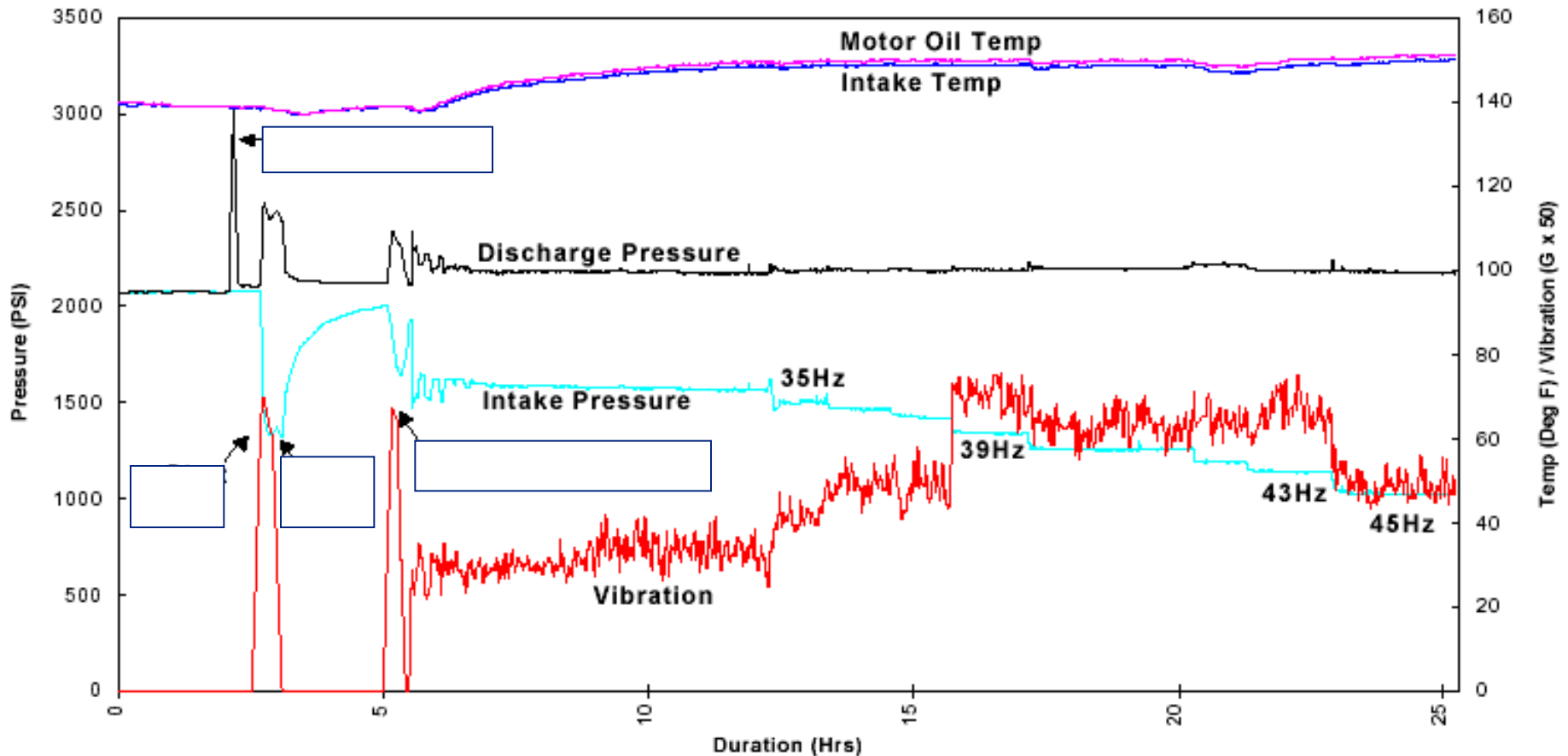
Measurements for ESP systems

What's measured	What it can help with
Pump discharge pressure	Water cut, tubing GOR estimation and well shut in protection
Well head pressure	Water cut and tubing GOR
Pump intake pressure	Gas locking (P_b and free gas volumes), reservoir drawdown, sand control, coning
ΔP_{pump}	Upthrust and downthrust protection; flow estimation, pump performance (wear, viscosity, materials)
Motor temperature	Overheating
Motor amps	Overload/underload protection
Current leakage	Overheating – insulation breakdown
Vibration	Fatigue, loading

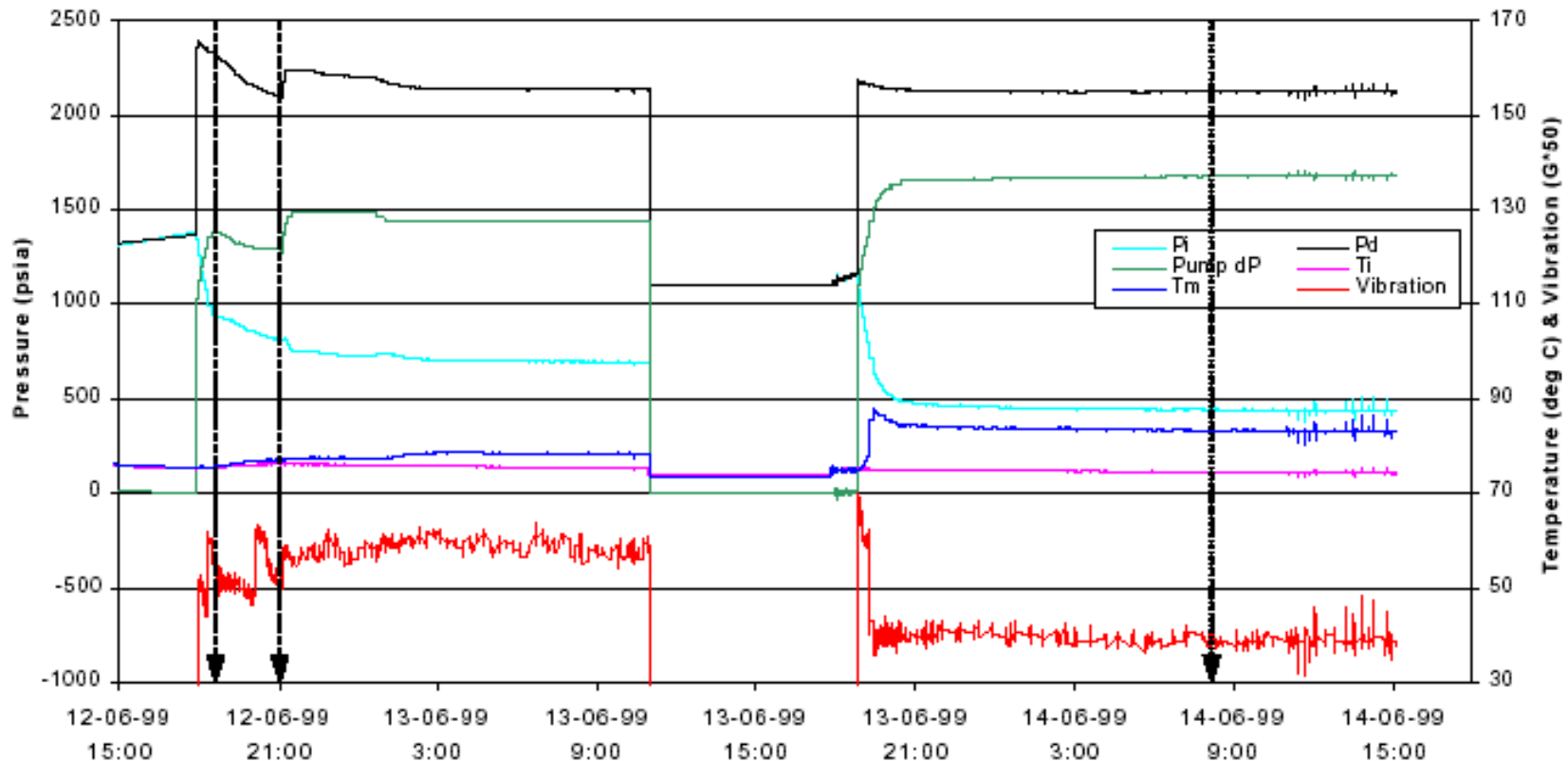
ESP diagnostics examples



ESP Diagnostic Examples



ESP diagnostics and Time slice analysis



ESP diagnostics and Time slice analysis

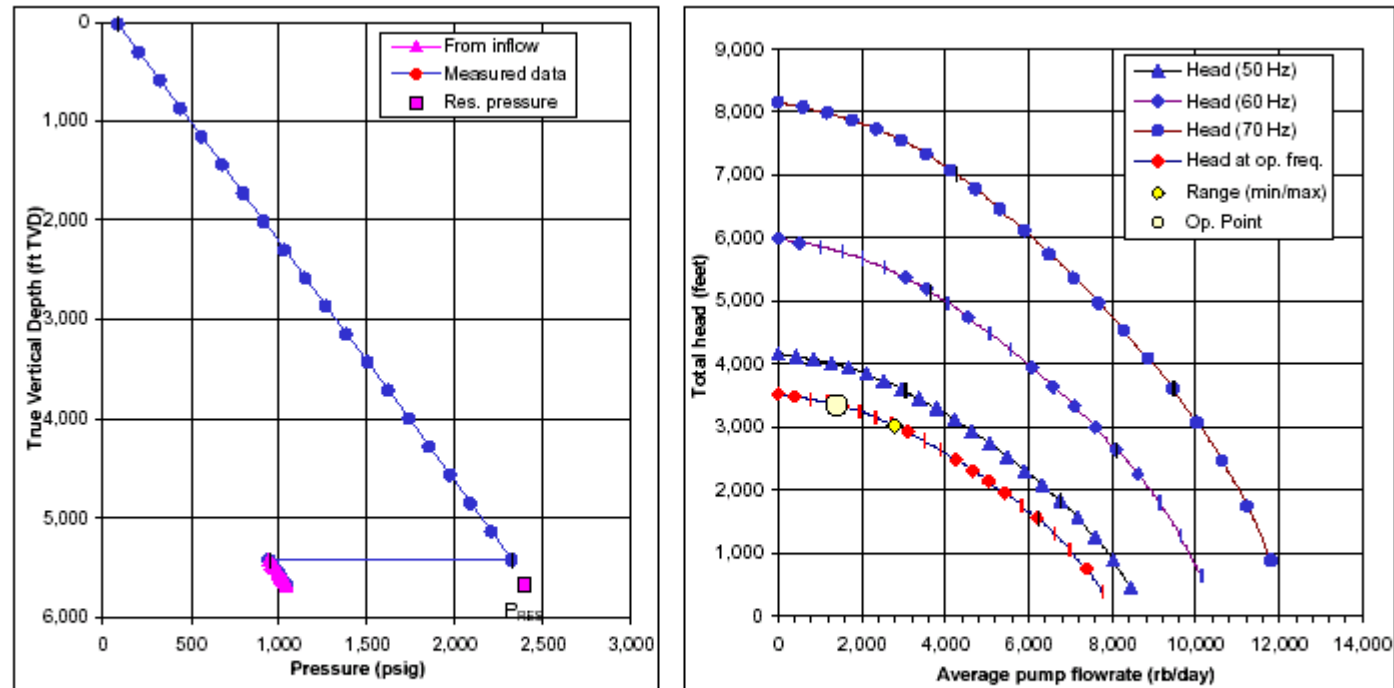


Figure 11 – Gradient traverse plot for Match #1.